Measuring Electromagnetic Induction

The Peak EMF Induced by a Magnet Falling Through a Coil

Chhaya Ramnani (u8336188)

Lab partners: Tsz Wing Candice Ng, Chinatsu Komada

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1.0 Abstract

This experiment examined how the peak electromotive force (emf) induced by a magnet falling through a coil depends on three things: the number of coil turns N, the coil radius R, and the magnet's speed s as it passed through. The magnet's dipole moment was first found from calibration measurements of its on-axis field, giving $m = (1.18 \pm 0.07)\ \mathrm{A\,m^2}$. This was used to build a theoretical prediction for the peak emf, which the measured data from each of the three sub-experiments was then compared against directly: no scaling factor was fitted anywhere. The peak emf rose roughly linearly with N and with s, and fell off roughly as R^{-2} with coil radius, matching the predicted shape $\varepsilon_{\text{peak}} \propto Ns/R^2$. None of the three datasets matched the absolute theoretical prediction within uncertainty, and the N data disagreed by far the most (a measured-to-theory ratio of 0.461 ± 0.003). This was mainly put down to the magnet's finite size breaking the point-dipole assumption the theory relies on, and, specifically for the N sub-experiment, to an unresolved mismatch between two setups that should have given the same reading.

2.0 Introduction

2.1 Aims

This experiment set out to see how the peak emf induced by a magnet falling through a coil depends on three parameters: the number of turns in the coil, N; the coil radius, R; and the magnet's speed as it passed through, s. A dipole model of the magnet was calibrated first from independent field measurements, and the resulting theoretical prediction was then compared directly against the measured emf for each sub-experiment.